

Homework (Habanero)

Q1. What is the greatest common factor of $12x$ and $18y$?

- (A) 2
- (B) 3
- (C) 6
- (D) $6x$
- (E) $6y$

Q2. Factor out the greatest common factor from $9a + 12b$

- (A) $3(3a + 4b)$
- (B) $3(a + 4b)$
- (C) $3(3a - 4b)$
- (D) $9(a + b)$
- (E) $12(a + b)$

Q3. A patient needs to take 15 pills of medication 3 times a day. What is the total number of pills taken in a day in terms of the greatest common factor?

- (A) $3 \cdot 5$
- (B) $3 \cdot 15$
- (C) $5 \cdot 9$
- (D) $3(5)$
- (E) $5(3)$

Q4. What is the factored form of $24x^2 + 16x$?

- (A) $8x(3x + 2)$
- (B) $8(3x^2 + 2x)$
- (C) $8x(2x + 3)$
- (D) $4x(6x + 4)$
- (E) $4(6x^2 + 4x)$

Q5. A biologist is studying the growth of two types of cells. The

growth rate of type A cells is 12 cells per hour and type B cells is 9 cells per hour. What is the greatest common factor of their growth rates?

- (A) 2
- (B) 3
- (C) 6
- (D) 9
- (E) 12

Q6. Factor out the greatest common factor from $20x^2y + 16xy^2$

- (A) $4xy(5x + 4y)$
- (B) $4x(5xy + 4y)$
- (C) $4y(5x^2 + 4x)$
- (D) $4xy(4x + 5y)$
- (E) $8xy(x + y)$

Q7. A doctor prescribes 18 pills of medicine 2 times a day to a patient. What is the total number of pills taken in a day in terms of the greatest common factor?

- (A) $2 \cdot 9$
- (B) $2 \cdot 18$
- (C) $9 \cdot 2$
- (D) $2(9)$
- (E) $9(2)$

Q8. What is the factored form of $18x^2 + 12x$?

- (A) $6x(3x + 2)$
- (B) $6(3x^2 + 2x)$
- (C) $6x(2x + 3)$
- (D) $3x(6x + 4)$
- (E) $3(6x^2 + 4x)$

Open Ended Questions (FRQ)

Q9. A researcher is analyzing the statistics of two types of medical treatments. The first treatment has a success rate of 12

Q10. Factor out the greatest common factor from the expression $24x^3 + 16x^2$. Show your work and explain your answer.

Chef's Tips

- Tip 1: Emphasize the importance of finding the greatest common factor
- Tip 2: Use real-world examples from healthcare and biology to illustrate the concept
- Tip 3: Provide additional practice problems for students who need extra help